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Platform for Integrated Continuous Bioprocessing

Continuous processing of 100 g of monoclonal antibody in 24 hours has been demonstrated using lab-scale equipment.

Jul 01, 2017 By Xhorxhi Gjoka, Rene Gantier, Mark Schofield
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production volumes.

At the same time, pressure to reduce the cost of biopharmaceuticals is also mounting. This pressure comes from new biosimilars entering the market and the growing numbers of drugs targeting the same diseases. Many leading monoclonal antibodies (mAbs), which make up the largest category of biologics, will see their patents expire in the next few years. As a result, the quantity of drug product needed to meet market demand can fluctuate significantly.

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There is, consequently, the need to achieve greater cost and time efficiencies, along with increased flexibility in biomanufacturing. Process intensification through integrated continuous manufacturing is a promising solution that is being championed by FDA to not only address these issues but also increase process robustness and product quality.

Integrated continuous bioprocesses consisting of connected continuous unit operations have a much smaller footprint, require fewer if any hold steps, allow the adoption of single-use systems, and enable accelerated development and scale-up. Initial capital outlays and ongoing operating expenses are generally reduced as well. Operating under optimum steady-state conditions holds the promise of consistent processing and greater product quality.

An Integrated Continuous 100 g/day Downstream Bioprocess Using Benchtop Equipment

Developing integrated bioprocesses is not a simple matter, however. There are numerous factors that must be considered. The operating parameters for each unit operation must be modified to realize the benefits of operating continuously. For instance, the continuous harvesting step in which the product is separated from spent cells determines the throughput for the remaining steps in an integrated downstream process. Generally, as the series of purification steps are completed, the process volumes decrease and product concentrations increase. This creates a challenge in terms of equipment sizing.

To demonstrate an integrated downstream bioprocess and the advantages of a continuous platform, the output of a 50 L fed-batch bioreactor was processed using the benchtop continuous platform to deliver 100 g of formulated mAb product within 24 hours.

For the cell culture process, a Chinese hamster ovary (CHO)-based cell line expressing humanised IgG1 monoclonal antibody that binds human epidermal growth factor receptor 2 HER2 was employed. The final titer was 4g/L with 25-30 x 10⁶ cells/mL and 65-85% viability.

The integrated unit operations included:

- Continuous clarification followed by depth/sterile filtration and continuous concentration
- Continuous Protein A chromatography
- Continuous virus inactivation followed by depth/sterile filtration
- Continuous polishing chromatography
- Continuous viral clearance
- Continuous final formulation, including continuous concentration, buffer exchange, and sterile filtration.

Continuous clarification and concentration

Continuous clarification was achieved using acoustic wave separation (AWS) technology (Cadence Acoustic Separator, Pall Life Sciences) where continuous removal of cells was achieved in a single-use, closed system without any detectable impact on either the cells or the product itself. Unlike depth filtration, the technology does not foul over time. It therefore enables significant reductions in depth filtration area and overall cost of goods for cell removal. AWS removes the bulk of the cells (80% or more). This provides a cost advantage in many scenarios by reducing the amount of depth filter surface area required for clarification. AWS also provides a path to clarifying unusually challenging feedstocks that require large depth filter surface areas to process.

In this study, stable operation of the AWS was demonstrated for approximately 14 hours with an overall continuous clarification yield of ≥85-

Current Issue

Bioreactors Redefine Upstream Productivity

New bioreactor designs, coupled with better media, process intensification and analytics, continue to improve upstream bioprocessing.



Critical Knowledge for Preparing Audits

Addressing data integrity, quality culture, aging facilities, investigations/corrective actions and preventive actions, and risk management is key when conducting audits, says Susan J. Schniepp, executive vice-president of post-approval pharma and distinguished fellow, Regulatory Compliance Associates.

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90%. The flow rate was maintained at 3.6 L/hr for a cell culture fluid with a titer of approximately 4 g/L, a cell density of 25-30 x 10⁶ cells/mL, and a cell viability of 65-85%.

The harvest cell culture fluid (HCCF) was then further clarified using an in-line depth filtration stage to remove finer particles. Depth filtration was achieved using PDH6 (3-15 µm) and PDD1 (0.2-2 µm) HP depth filter capsules in sequence. Sterile filtration was then conducted using a filter capsule (375 cm²) (Supor EKV Kleenpak, Pall Life Sciences) prior to continuous concentration.

Continuous concentration was not employed in processing this harvest because the titer was relatively high at 4 g/L. For other processes with titers less than 2 g/L, continuous concentration was employed (Cadence Inline Concentrator, Pall Life Sciences). At these titers (<2 g/L), performance maps were generated to explore the feasibility of integration with the prior continuous clarification/filtration/sterilization and post continuous chromatography steps. Different process control strategies were also investigated to eliminate process transients. Ultimately, the process was optimized for robust integration and reproducibility.

Continuous chromatography and virus inactivation

Application of a continuous, countercurrent multi-column process development chromatography system (Cadence BioSMB Process Development [PD], Pall Life Sciences) allowed for an overall reduction in the amount of sorbent required for a process. By employing multiple columns and loading columns in series, it is possible to improve the useable capacity of chromatography sorbents. Loading multiple columns in series also enables operation at high flow rates/short residence times. This allows for more rapid purification cycles and a reduction in sorbent volume (and therefore cost). For capture chromatography, reducing the amount of Protein A sorbent required is particularly desirable since Protein A sorbent can be 3-5 times more expensive than non-affinity sorbents, such as ion exchangers.

And, because achieving the highest possible purity is the goal of the polishing steps, the greater capacity afforded by continuous processing generally allows for operation under more stringent conditions to maximize purity levels. In the context of the integrated platform, it is also challenging to create matching unit operations. Each chromatography step must be designed with the previous and following operations in mind; specifically, the throughput and cycle times must be considered to ensure that the flow rates in and out of the chromatography operation match the other steps in the overall process.

In the current study, the fluid obtained following clarification was subjected to capture chromatography using Protein A (KanCapA, KANEKA) as the sorbent. The optimized multi-column chromatography method took into consideration the maximum linear velocity of the sorbent (500cm/hr), the flow rate from clarification (50mL/min), the titer (4g/L), the duration of the non-load steps, the availability of different column inner diameters at each scale and the need to produce at least 100 g/day in a 24-hour period. An 8-column process (50 mL each), with 4 columns in the load zone configured with 2-in-series and 2-in-parallel, was selected.

The eluate from the capture column was then subjected to virus inactivation (VI). A prototype VI process was performed using the multi-column process development chromatography system. Integrated continuous capture and VI was operated for approximately 14 hours (>20 cycles). In line analytics, UV, conductivity and pH show that the process was consistent and stable over this time. After VI, the product was depth and sterile filtered to remove any particulates before going to the polishing steps and sterile filters.

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Keeping Aggregation Under Control

Characterizing and controlling protein aggregation is vital to ensure safety and efficacy of a biopharmaceutical product. In this interview, important aspects of protein aggregation and the tools available to address this issue are discussed.

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Ensuring Smooth Tech Transfer of Bioprocess Operations

Experience, communication, collaboration, transparency, planning, and prioritization contribute to success.



Continuous polishing was achieved using an anion exchange membrane (AEX) (Mustang Q, Pall Life Sciences) followed by a mixed-mode cation exchange (MMCEX) sorbent (CMM HyperCel, Pall Life Sciences).

The AEX and MMCEX processes were integrated and performed on one multi-column process development chromatography system. As AEX is operated in flow-through mode, it has a much higher capacity (>3 g/mL) than the MMCEX, which is operated in bind and elute mode (>40 mg/mL). Therefore, integrating the steps creates a challenge. For the AEX, a process with three devices was selected. This allowed one device to always be loaded, while conferring enough time after loading for the device to be regenerated for >1 hour (1 N NaOH) before being loaded again. As the capacity for the MMCEX in bind and elute mode is lower than AEX, the MMCEX cycle is much shorter. In fact, it is roughly a third of the AEX cycle time. As a result, the MMCEX method, which contains six columns (two loaded, four performing non-load steps), completes one cycle in the time it takes to load one AEX membrane. Integration of the different flow rates across all the columns and organization of the loading and non-loading operations is crucial.

Notably, this order of unit operations (AEX followed by MMCEX) reverses the typical order used in traditional mAb processes. However, this order was selected to avoid the need for buffer exchange between the two steps and to streamline continuous operation. The AEX sorbent conducts the best host cell protein (HCP) reduction at high pH and low conductivity. The mixed-mode sorbent can be loaded at these conditions (pH8, and conductivity of 6mS/cm) at relatively high capacities >40mg/m. Importantly, similar product quality—~1.3% aggregates and greater than 60-fold HCP reduction—was achieved, regardless of the sequence. The polishing process in the optimum order was then run for more than 30 cycles with stable performance: <20 ppm HCP and <2% aggregates.

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